

AYAAN KHAN

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EDUCATION

National University of Computer and Emerging Sciences (FAST-NUCES)
Bachelor of Science, Computer Science

Islamabad

Aug 2022 - June 2026

Semester 8

Relevant Coursework: Deep Learning, Generative AI, DevOps, Cloud Computing, Software Design and Analysis, Software Engineering, Data Structures, Computer Networks, Operating Systems, Database Systems,

EXPERIENCE

Kodify

Islamabad

Project Associate

Apr 2026 – Present

- Improved cross-team coordination through Jira workflow management and project documentation.
- Accelerated feature planning by drafting PRDs, UATs, training manuals, and technical reports.
- Standardized software and hardware documentation by restructuring functional requirements.

sīParadigm Diagnostic Informatics

Islamabad

Data Analyst Intern

Jul 2025 – Aug 2025

- Automated preprocessing of 200+ GB medical PDFs into AI-ready datasets (Python, PostgreSQL).
- Applied spaCy NER for metadata matching, improving record linkage accuracy to 90%+.
- Built ETL & Excel QA tooling (Pandas) cutting manual effort by ~70%.

DreamAI Software

Islamabad (Remote)

Agentic AI Intern

Jul 2025 – Aug 2025

- Built multi-agent weather chatbot (Google ADK, Gemini API, WeatherAPI) for real-time forecasts.
- Designed evaluation loops (OpenAI Evals, Pydantic AI) for agent testing and integration.

Teresol Pvt. Ltd.

Islamabad

Software Development Intern

Jul 2023 – Aug 2023

- Developed/debugged C/C++ modules for embedded Linux
- automated workflows with Bash scripting .

SELECTED PROJECTS

Pose2Play - VR-Based Adaptive Learning for Gamified Rehabilitation (Computer Vision, LSTM, PyTorch, MediaPipe, Flask, Unity)

- Worked on a computer vision model using MediaPipe to extract joint angles for rehabilitation exercises in real-time.
- Trained an LSTM on multiple datasets to assess movement quality from temporal joint-angle sequences and power adaptive form feedback.
- Established a personalized adaptive learning engine using a Reinforcement Learning model that updates target joint angles based on user performance trends across repetitions.
- Created a VR environment via Unity with a gamification system to enhance user engagement during therapy.

Event Booking Microservices Platform (FastAPI, Javascript, Flask, Docker, Kubernetes, Terraform, Argo CD)

- Designed and deployed a scalable event booking system using containerized microservices on AWS.
- Implemented GitOps-driven CI/CD with Argo CD and GitHub Actions, improving deployment reliability with automated testing.
- Integrated RabbitMQ for inter-service communication, ensuring fault-tolerant operations.

Financial Sentiment Analysis with RAG (FinBERT, Flan-T5, FAISS, PyTorch, HuggingFace)

- Achieved 96.29% sentiment accuracy by fine-tuning FinBERT and Flan-T5 on financial text data.
- Improved contextual retrieval by implementing a FAISS based RAG pipeline over financial documents.
- Enabled real-time sentiment analysis by deploying the model through a lightweight inference API.

SKILLS

Languages: Python, C++, JavaScript, SQL

AI/ML: PyTorch, Transformers, HuggingFace, RAG, NLP, Computer Vision, LLM Finetuning, Agentic AI

Backend: FastAPI, Flask, REST APIs, Microservices, MERN

DevOps/Cloud: Docker, Kubernetes, Terraform, GitHub Actions, Argo CD, AWS

Databases: PostgreSQL, MongoDB

LEADERSHIP AND HONORS

Top 50 in Pakistan — National Mathematics Olympiad Camp (COMSATS Lahore)

Dean's List — Fall 2025 (Semester 7)

Secretary General, FAST Debating Society (2024–25) — Led operations, workshops, inter-university events.

President, Potohar MUN (2021–22) — 300+ delegates, managed PKR 1.78M budget, secured PKR 400K sponsorships.